

Gaurav Gupta

SCIENTIST/ENGINEER-SC

Space Application Centre, Ahmedabad

☎ (+91) 9083722796 | ✉ gauravg.aero@gmail.com | 🏠 <https://gauravity30.github.io/> | 📺 gauravity30 | 📺 gauravgupta030 | 📧 Gaurav Gupta

Summary

Fluid dynamicist specializing in computational studies of biological and bio-inspired flow problems. Primary experience encompasses investigating avian flight aerodynamics, modeling microgravity two-phase flows, and developing custom flow-solver and post-processing subroutines in Fortran and Python respectively. Currently working on capillary-driven flows and interfacial phenomena under microgravity at ISRO.

Research Interests: Biological fluid mechanics; animal locomotion; low-Reynolds number aerodynamics; vortex dynamics; fluid-structure interactions; turbulent and interfacial flows.

Research Experience

Space Application Centre, Indian Space Research Organization (ISRO)

Ahmedabad, Gujarat

SCIENTIST/ENGINEER - SC

Jul. 2025 - Present

Microgravity Fluid Mechanics

- Formulated the conceptual system architecture for a space-based recirculating aquatic life support payload.
- Conducted **two-phase simulations** to investigate the **capillary-driven fluid flow and transport** under microgravity using OpenFOAM.
- Currently, investigating stability and dynamics of an **air-water interface** under microgravity using OpenFOAM.

Opto-Mechanical Engineering for Opto-Quantum Payloads

- Developed space-qualified opto-mechanical systems and automated CAD/CAE workflows using Siemens NX Open

Indian Institute of Technology, Kharagpur

Kharagpur, India

RESEARCH INTERN (SUPERVISORS: DR. SANDEEP SAHA AND DR. MANU K. V.)

Mar. 2023 - May. 2025

- Investigated **avian-tail aerodynamics** in gliding flight through **high-fidelity computational methods**, with emphasis on assessing the influence of tail geometry.
- Developed a representative common swift geometry based on the experimental configuration of P.Henningsson et al. (2011) to enable comparison with the published data.
- Constructed a structured mesh of the common swift geometry using multi-block topology in Pointwise
- Investigated high-fidelity computational approaches, evaluating the suitability of **DNS**, **RANS-assisted DNS**, **DES** and **RANS** methods for resolving flow features and wake vortices at relevant Re .
- Conducted **2D DNS** (using NEK5000), and **3D RANS** (using SU2) of the representative common swift geometry at $Re = 10000 - 20000$.

Education

Indian Institute of Space Science and Technology

Thiruvananthapuram, India

B.TECH. IN AEROSPACE ENGINEERING

Dec. 2021 - Jun. 2025

- Cumulative Grade Point Average (CGPA): 8.28/10
- Relevant coursework: Aerodynamics, Computational Fluid Dynamics, Aircraft Design, Boundary Layer Theory, Finite Element Method, Flight Dynamics & Control, Introduction to Astrobiology

Scientific Software & Open-Source Contributions

Incompact3D

CFD, Fortran

HIGH-ORDER FINITE DIFFERENCE FLOW SOLVER



- Developed and integrated a Fortran subroutine calculating aerodynamic forces from **3D immersed body simulations** in Incompact3D.
- Working on improving the performance of importing STL files for high-fidelity simulations using Incompact3D.

PyMech

CFD, Python, NEK5000

A PYTHON SOFTWARE SUITE FOR NEK5000 AND SIMSON



- Improving the in-built open_dataset function to support unstructured datasets produced using NEK5000.
- Used for post-processing of high-fidelity simulation data using NEK5000.

TurbCourse

CFD, Python, NEK5000

A COURSE IN TURBULENCE SIMULATION



- Developed and optimized **POD** and **DMD** modules for extracting coherent structures and frequency data from high-fidelity simulations.
- Uses PyMech for reading/writing of NEK5000 data.



- An incompressible inviscid flow solver based on **Artificial Compressibility Method** was developed using FreeFEM++.
- The code is capable of solving 2D problems based on in-built FreeFEM++ and GMSH meshes.

Publications

RESEARCH ARTICLES

- [1] **Gaurav Gupta** and Md Laraib Salam, "Determining surface tension of various liquids and shear modulus of paper using crumpling effect," *Journal of Emerging Investigators*, vol. 4, 2021. doi: **10.59720/21-104**
- [2] **Gaurav Gupta**, Laraib Salam, and Arunabha Adhikari, "Formation and sticking of air bubbles in water in d-block containers," *Journal of Emerging Investigators*, vol. 4, 2021. doi: **10.59720/21-010**

CONFERENCE/TECHNICAL PROCEEDINGS

- [3] **Gaurav Gupta** and Ankit Sharma, "Enabling automation in opto-quantum communication program through APIs," in *ISRO/DOS Seminar on Computer Technology, Cyber Security and Mission IT*, Internal proceedings; Non-Public, 2026.

Honors

Scientist of the Future 2020

Moscow State University (Online)

4TH PLACE (PHYSICS AND ASTRONOMY)

2020

- Research project titled *Formation and sticking of air bubbles in water in d-block containers*.
- Subsequently published in the *Journal of Emerging Investigators*.

Previous Graduate Admissions

Oxford University

D.PHIL IN BIOLOGY, OXFORD FLIGHT GROUP

2025

- Offered admission under the guidance of Prof. Graham Taylor; deferred to accept Scientist/Engineer role at ISRO.

Johns Hopkins University

M.S. IN ENGINEERING

2025

- Offered admission; deferred to accept Scientist/Engineer role at ISRO.

Research Methods

Flow Physics	Unsteady aerodynamics, multiphase and interfacial flows, fluid-structure interaction
Numerical Methods	DNS, RANS-assisted DNS, DES, immersed-boundary methods
Data Analysis	POD, DMD and frequency analysis
Scientific Computing	Parallel computing, numerical simulations and scientific programming

Technical Skills

Solvers	Incompact3D, NEK5000, SU2, NEKTAR++, OpenFOAM
Meshing/Post-Processing	GMSH, Pointwise, ParaView, VisIt
Programming	C++, Fortran, FreeFEM++, Julia, Python, MATLAB
Tools	Linux, Git, LaTeX
CAD/CAE	Siemens NX, Solidworks

Selected Certificates & Examinations

Mar 2025 **Graduate Aptitude Test - Aerospace Engineering**, Indian Institute of Technology, Roorkee

All India Rank 60

Apr 2024 **High Performance Computing**, IIT Bombay and NPTEL

Extracurricular

- Directed IIST's technical festival (Conscientia) for 1000+ participants as Chief Coordinator, managing a INR 2.2 million budget and securing an INR 400,000 profit.
- Led team initiatives in designing fixed-wing and rotor aircraft as Student Coordinator, while orchestrating UAV design workshops and competitions.